

HVIA TASK OLIST BUSINESS DISCOVERY

Link website: [Sobre a Olist: Nossa missão é fortalecer o comércio no mundo](#)

Olist is a leading Brazilian e-commerce platform and a 'unicorn' company.

-The main things the platform does and is known for:

1-Basic Business Model (How does it work?)

Olist works as an intermediary or a multi-department store within major marketplaces (Marketplace Hub). Instead of a small merchant setting up their own store and contracting with each platform individually, they register with Olist under a single contract. Then, Olist takes care of listing their products and selling them under the name "Olist Store" on major global and Brazilian e-commerce platforms (like Amazon, Mercado Livre, Shopee, and Magalu).

2-The services and solutions

Olist offers have evolved from just being a connecting platform to becoming a full-fledged commerce operating system, including: Company Management System (ERP): for managing inventory, invoices, and sales all in one place (like their Tiny system). Logistics and Shipping Solutions (Olist Envios): to provide fast and affordable shipping and delivery options through their logistics partners. AI Tools (Lis): to help merchants with product pricing, automatic invoicing, and preventing stockouts. Financial Services: providing payment systems and credit facilities for merchants.

-Business model in 2016 to 2018:

Olist's business model was entirely based on an innovative strategy known as 'Store-in-Store' or 'Marketplace Aggregator'. In those years, the company hadn't yet expanded into acquiring the major logistics software companies it now owns, and its business model was based on three main pillars as a service software company (B2B SaaS):

1-How did the business model work?:

Olist used to work as middleware software connecting small merchants to big platforms.

- Instead of the merchant opening an account on Amazon, Mercado Livre, or Submarino, they would just sign up on Olist.
- Olist would take the merchant's products and display them in the big stores under its unified store name, 'Olist Store.'
- When the end customer buys from Amazon (for example), the invoice goes to Olist, and then the platform sends an automatic notification to the original merchant to package and ship the product.

2-How was Olist making money during this period?

- Fixed Monthly Subscription (SaaS Subscription): The small merchant pays a monthly fee to use the platform's software to manage their inventory and products.
- Commission on Sales: Olist used to take a percentage (commission) from each successful sale made through the 'Olist Store' on major platforms.

3-What was the competitive advantage that she sold to the merchants back then?

- Reputation Sharing: Big platforms give priority in visibility to stores with high ratings. Small merchants were taking advantage of Olist Store's very high rating to sell immediately without needing to build a reputation from scratch.
- One contract for all platforms (Single Contract): Saves costs and avoids the complicated contracting and technical integration with each platform separately.
- Discounted shipping (Logistics Partnering): Connecting small merchants to very competitive shipping rates through Olist's partnerships with the Brazilian Post and shipping companies, to reduce the shipping (freight) cost for the consumer.

Link data: [Brazilian E-Commerce Public Dataset by Olist](#)

Detailed table (Foreign Keys):

1-Customers and Orders:

orders.customer_id = customers.customer_id

2-Orders and Details (Items, Payments, and Reviews):

order_items.order_id = orders.order_id

order_payments.order_id = orders.order_id

order_reviews.order_id = orders.order_id

3-Products and sellers (linked to the order_items tables):

order_items.product_id = products.product_id

order_items.seller_id = sellers.seller_id

4-Category name translation (changing the category name from Portuguese to English):

products.product_category_name = product_category_name_translation.product_category_name

1. Financial Performance & Purchasing Behavior

-Sales Volume & Order Value: Total platform sales reached approximately BRL 15.42 million, with an average order value of BRL 159.86.

-Top Product Categories: The highest-volume product categories are bed & bath/table, health & beauty, sports & leisure, furniture & decor, and computers & accessories.

-Sales Trends: The platform experienced steady growth from early 2017 through mid-2018, with a pronounced sales spike in November 2017.

-Payment Methods & Installments: Credit cards have the highest average payment value (approximately BRL 163.94) and meaningful installment usage, averaging 3.51 installments.OV.

2. Geographic Distribution & Supply Chain Dynamics

-Regional Concentration: Sales are overwhelmingly concentrated in Brazil's Southeastern region. São Paulo (SP) leads the platform by accounting for over **41.98%** of total orders, followed by Rio de Janeiro (**12.8%**) and Minas Gerais (**11.77%**).

-Intrastate vs. Interstate Shipping:

Within the same state: Deliveries take only **5 to 7 days** on average, maintaining a negligible delay rate of **5.99%**.

Across state lines: Average delivery times rise to **15 to 18 days**, with delay rates exceeding **9.16%**.

-Logistical Bottlenecks: Delivery performance varies significantly by state. Alagoas (AL) recorded the highest delay rate at **23.93%**, followed by Maranhão (MA) at **19.67%**, while Piauí (PI), Ceará (CE), and Sergipe (SE) also exceeded **15%**.

3. Shipping Delays & Impact on Customer Satisfaction

-Overall Delivery Timelines: Average delivery time was **12.56 days**, while **8.11%** of delivered orders missed their estimated delivery date.

-The Delay-Rating Correlation: On-time orders received an average rating of **4.29/5**, compared with **2.57/5** for delayed orders.

-Core Customer Pain Points: Low ratings (1 or 2 stars) represent **12.76 %** of all platform reviews, with 1-star ratings alone accounting for **9.72%**. These low scores stem from three primary causes:

Courier and transit delays.

Slow seller handling/processing time before dispatching to the carrier.

Product quality issues or descriptions not matching the delivered item.

4. Seller Performance & Operational Risk

-High-Volume Sellers & Ratings: Top-earning sellers (generating 200,000–300,000+ BRL) mostly operate in electronics, watches, and auto parts. However, **sales volume does not correlate with high ratings**. Some top sellers average only 3.7 to 3.9 out of 5, as high order volumes strain their fulfillment pipelines and increase packaging or shipping errors.

-High-Risk, High-Volume Sellers: One high-volume seller met the defined risk criteria, handling **973** orders with approximately BRL 237.8K in revenue, a **10.07%** delay rate, and a **3.50/5** average rating. This indicates a meaningful customer-experience risk at high sales volume. .

-Performance Variations by Location & Product Type:

Seller Location: Sellers in São Paulo achieve the best operational metrics and lowest delay rates due to their proximity to central logistics hubs, whereas sellers in states like Rio de Janeiro experience noticeably higher delay rates.

Product Type: Product weight and volume showed only weak linear relationships with delivery delay rate, suggesting that geographic and shipping-related factors may be more actionable drivers than product size alone.

Based on all the analyses and numbers we pulled from the Olist dataset:

Observation 1: Shipping & Delivery Delays

Data Finding: 8.11% of delivered orders missed their estimated delivery date. On-time orders averaged 4.29/5, compared with 2.57/5 for delayed orders. Interstate shipments also showed substantially longer delivery times and higher delay rates than intrastate shipments.

Business Impact: High customer churn as negative experiences stop repeat purchases, explaining low customer retention. Severe ratings also damage Olist's brand reputation and inflate customer service support costs.

Proposed Action: Implement a **Predictive Dynamic ETA Model** using machine learning to calculate accurate delivery dates based on customer location, seller origin, and product dimensions rather than relying on static estimates. Establish an **Early-Warning Delay System** to detect high-risk delays in advance, notify logistics partners, and proactively inform customers to manage expectations.

Observation 2: Seller Performance & Operational Risk

Data Finding: We identified one high-volume seller meeting the defined risk criteria, handling 973 orders with approximately BRL 237.8K in revenue, a 10.07% delay rate, and a 3.50/5 average rating.

Business Impact: High-volume sellers with poor operational performance disproportionately harm Olist's brand value, as their large sales volume affects thousands of customers daily and erodes trust across partner marketplaces.

Proposed Action: Deploy a **Predictive Seller Risk Scoring** model to continuously evaluate seller metrics, flagging underperforming accounts to account management teams for intervention or temporary daily order caps until preparation standards improve. Enforce strict **Service Level Agreements (SLAs)** with defined time windows for order dispatch.

Observation 3: Customer Feedback & Sentiment Analysis

Data Finding: Approximately 11% to 14% of all customer ratings are low scores (1 or 2 stars), the vast majority of which contain qualitative text feedback and complaints written in Portuguese.

Business Impact: Failing to systematically analyze these textual complaints prevents the company from identifying the precise root causes behind customer dissatisfaction—such as product damage, carrier delays, or unresponsive sellers.

Proposed Action: NLP-based sentiment and topic classification model to automatically process feedback text, categorize reported issues, and immediately flag high-risk unhappy customers for priority outreach by customer support for proactive churn prevention.

Outreach Message Draft

Dear [Decision Maker's Name / Title: e.g., Head of Customer Experience / Logistics],

I have been analyzing Olist's public e-commerce order dataset and identified two critical operational friction points that are directly suppressing repeat purchases and overall revenue growth:

1. Delayed Deliveries

Current Impact (Before Solution):

Delivery delays affect **8.11%** of delivered orders, while average ratings fall from **4.29/5** for on-time orders to **2.57/5** for delayed orders. Interstate shipments also take substantially longer and experience higher delay rates than intrastate shipments, creating a clear opportunity for predictive ETA and early-warning intervention.

Proposed Solution:

AI-Powered Dynamic ETA & Delay Early-Warning System: Developing a predictive machine learning model that calculates accurate, real-time arrival estimates based on shipping geography, courier performance, and seller dispatch speeds, while sending early warnings to logistics teams for high-risk orders.

Financial Impact (After Solution):

Reduce delivery delays, improve customer satisfaction, and provide earlier intervention for high-risk orders.

2. Unhandled Negative Reviews & Customer Sentiment

Current Impact (Before Solution):

Low-score reviews (1 and 2 stars) account for **14% to 15% of all platform feedback**. The vast majority contain detailed text descriptions detailing product defects, merchant errors, or delivery issues, yet lack an automated, real-time mechanism to intervene and recover unhappy buyers.

Proposed Solution:

NLP Automated Sentiment & Churn Prevention Engine: Implementing a Natural Language Processing (NLP) pipeline that instantly analyzes review text upon submission, categorizes root causes (seller vs. carrier vs. product quality), and flags high-risk churn cases directly to customer support for proactive recovery.

Financial Impact (After Solution):

Increases customer retention, boosts Customer Lifetime Value (LTV), and recovers sales that would otherwise be lost to competing e-commerce platforms.

At **HVIA Data & AI Solutions**, we specialize in transforming raw operational data into measurable financial growth.

Would you be open to a brief **15-minute discovery call** to review these data findings in detail and explore how we can implement these solutions for Olist?

Best regards,

Mahmoud Abdelrazek

Data & AI Solutions Specialist | HVIA